

SID Methods and Synthetic Experiments

A simple guide to what we estimated, how we tested it, and what actually worked

Internal teaching note for the diffusionCircuit project

13 July 2026

Bottom line. The clearest success is not “diffusion SID.” It is *contrast-aligned SID*: compare two declared histories directly, then estimate either their density ratio, a balanced endpoint classifier, or a typed moment difference. This works on location, covariance, separated mixture modes, and the hard fourth-order G8 path effect. A ratio critic is the best current clean infinitesimal estimator. Diffusion and flow are useful predictive-law backends and sample generators, but their native training objectives do not by themselves recover the history effect we care about.

If you remember only five things:

1. We observe a history vector $H = h$ and a future response or path Y . The basic object is the conditional predictive law $P_h = P(Y \mid H = h)$.
2. A good model of P_h is not automatically a good model of how P_h changes with history.
3. The clean tangent asks for an infinitesimal derivative. The finite method asks for the difference between two supported histories h_- and h_+ .
4. $\text{NRMSE} = 0$ is perfect; $\text{NRMSE} = 1$ is the zero-effect predictor. Above one is worse than saying “nothing changes.”
5. On the simple tests, G1, G2, and G4 work; G3 does not. On G8, direct contrast classification works while generic full-law generators largely erase the target effect.

This document uses only frozen July 13 results. No model was retrained for this guide. Real neural data are intentionally out of scope.

1 The problem, stated slowly

Let h summarize the observed past and let Y be the next observation or a future path. Every history h selects a probability law

$$P_h(A) = \Pr(Y \in A \mid H = h), \quad p_h(y) = p(y \mid h).$$

There are four different questions in the experiments. They should not be treated as interchangeable.

Question	Quantity	What a successful answer means
Predictive-law fit	$p_h(y)$ or samples from P_h	Future observations receive high likelihood and generated samples have the right distribution.
Outcome score	$\nabla_y \log p_{h,\sigma}(y)$	The model points toward high-density responses at a chosen noise level σ .
Clean history tangent	$\ell_v(y; h) = D_v \log p_h(y)$	The model recovers how log density changes under an infinitesimal history perturbation in direction v .
Finite typed effect	$\Delta_{\phi,\delta}(h, v)$ below	The model recovers how a declared feature—mean, covariance, skew, occupancy, or a path motif—changes between two nearby supported histories.

For the finite comparison, define

$$h_- = h - \delta v, \quad h_+ = h + \delta v,$$

and let p_- and p_+ be their predictive densities. The bounded finite witness is

$$W_\delta(y; h, v) = \frac{p_+(y) - p_-(y)}{\delta\{p_+(y) + p_-(y)\}} = \frac{1}{\delta} \tanh \left\{ \frac{\log p_+(y) - \log p_-(y)}{2} \right\}.$$

It says *which outcomes become more or less likely*. For a scientifically chosen response feature $\phi(Y)$, the typed effect is simply

$$\Delta_{\phi,\delta}(h, v) = \frac{\mathbb{E}_{P_+}[\phi(Y)] - \mathbb{E}_{P_-}[\phi(Y)]}{2\delta}.$$

Examples are $\phi(Y) = Y$ for a mean effect, $YY^\top$ for covariance, a cubic feature for skew, a mode indicator for occupancy, or a fourth-order path feature for G8.

Bottom line. The witness is a full description of the finite law change. A typed effect is one projection of that change. If the scientific question is already typed, estimating the typed effect directly is usually easier than reconstructing the entire witness.

Why the word “score” caused confusion

Diffusion learns an *outcome* score $\nabla_y \log p_{h,\sigma}(y)$. SID asks for a *history* derivative $D_v \log p_h(y)$. SBTG-like mixed operators involve $\nabla_y D_v \log p_h(y)$. These objects are related, but none is numerically equal to the others. Moving from one to another requires differentiation, centering, or an inverse reconstruction problem. Therefore low diffusion score error does not imply low SID error; the experiments verify this directly.

2 The methods: first separate the backend from the SID adapter

Most of the apparent methods are combinations of two choices:

$$\boxed{\text{data } (h, y)} \longrightarrow \boxed{\text{predictive-law backend}} \longrightarrow \boxed{\text{SID adapter}} \longrightarrow \boxed{\text{typed readout } \phi}.$$

A backend represents P_h . An adapter compares histories. A readout declares what kind of change matters. A diffusion model with a coupled-sample adapter and a mean readout is therefore a different estimator from the same diffusion model with a score-derivative adapter.

2.1 Predictive-law backends

Backend	Training objective	Representation returned	Main limitation
Mean MLP	Mean squared error	$\hat{\mu}(h)$	Sees only conditional mean.
Gaussian NLL	Exact Gaussian negative log likelihood	Diagonal $\hat{\mu}(h), \hat{\sigma}^2(h)$; normalized density and samples	Cannot represent multimodality or rotated residual covariance.
Constrained Gaussian DSM	Multi-noise denoising score loss, but constrained to a positive Gaussian variance	The same Gaussian law via a score objective	Objective control, not a more flexible law.
Autoregressive MDN	Exact NLL for eight Gaussian components per response coordinate	Normalized flexible density, score, and ancestral samples	Coordinate ordering and sequential sampling.
Autoregressive Transformer	Exact mixture NLL with causal response tokens	More flexible normalized density for high-dimensional paths	Expensive and still may ignore weak history-controlled mixture weights.
Affine flow	Exact change-of-variables NLL with six conditional coupling layers	Normalized density, score, and samples	A simple invertible geometry can miss disconnected modes.
EDM diffusion	Weighted denoising regression across noise scales	Noisy score, denoiser, and iterative samples	Native loss targets response denoising, not history SID.
Gaussian-anchored EDM	EDM residual around a learned Gaussian plus an NLL term	Stabilized sampler with location/scale baseline	Still inherits the objective mismatch.
Flow matching	Velocity regression along a Gaussian-to-data interpolation	ODE sampler	No normalized density in this implementation; history effect comes only through a separate adapter.
Gaussian-source flow	Flow matching from a learned conditional Gaussian source	Stabilized coupled samples	Can preserve easy moments while missing higher-order mass.
Bounded energy / interpolant	Bounded density tilt or fixed-anchor transport regression	Alternative normalized or generative routes	Stable sampling did not rescue the hard G8 effect.

All neural backends in the frozen core used standardized inputs and responses. Most MLPs had two hidden layers of width 56–64. EDM and flow used 18 Heun integration steps. These were intentionally compact CPU models, not frontier-scale generators.

2.2 SID adapters

1. Autodifferentiate a normalized density. If a backend returns $\log \hat{p}(y | h)$, differentiate it with respect to h to estimate the clean tangent, or evaluate it at h_+ and h_- to form the exact bounded finite witness. This is principled but inherits every error in the fitted density and its history derivative.

2. Ratio critic for the clean tangent. Train a classifier to distinguish matched pairs (h_i, y_i) from shuffled pairs $(h_i, y_{\pi(i)})$. At the population optimum its logit is

$$r(h, y) = \log \frac{p(h, y)}{p(h)p(y)} = \log p(y | h) - \log p(y).$$

Because the final term is independent of h , $D_v r(h, y) = D_v \log p(y | h)$. The implementation used rank-32 learned history and response features, their bilinear interaction, binary logistic loss, and automatic differentiation with respect to history. It does *not* provide a normalized density or samples.

3. Balanced finite classifier. Draw equal numbers of endpoint samples $Y_+ \sim P_{h_+}$ and $Y_- \sim P_{h_-}$, label them $+$ and $-$, and fit a calibrated classifier $\hat{\eta}(y, h) = \Pr(+ | y, h)$. Equal class priors give

$$\eta(y, h) = \frac{p_+(y)}{p_+(y) + p_-(y)}, \quad W_\delta(y; h, v) = \frac{2\eta(y, h) - 1}{\delta}.$$

The frozen finite panel used a two-layer width-64 MLP, binary cross-entropy, early stopping, then temperature-and-bias calibration on an independent balanced split. The classifier received the unperturbed anchor history and response, but not the endpoint sign as an input feature.

4. Direct coupled samples. Generate from h_+ and h_- with the same base randomness, then compute the difference of $\phi(Y_+)$ and $\phi(Y_-)$. Common random numbers reduce variance. This is often the best route when only a typed effect is needed; it never reconstructs a full witness.

5. Signed Riesz dictionary. Project the witness onto a fixed dictionary of linear, quadratic, cross, cubic, and smooth-tail response features. The coefficients solve a ridge-regularized moment equation. This was meant to estimate difficult channels without a full density, but the tested dictionary was generally unstable or blind; it should not be treated as a current success.

6. Score-derived and guided adapters. Diffusion score differences, model-sample centering, stochastic interpolation, and classifier guidance all try to reuse a generative backend. They are useful diagnostic bridges. None currently matches the reliability of direct ratios/classifiers on the hard shape effects.

Bottom line. The two most developed SID estimators are the ratio critic for a clean local tangent and the balanced endpoint classifier/direct typed contrast for a finite effect. Diffusion, MDN, flow, and Gaussian models are backends that may feed those adapters.

3 The four simplest synthetic systems

The first four systems were chosen so that each changes a different aspect of the predictive law. All have an 8-dimensional history. G1, G2, and G4 have an 8-dimensional response in the frozen core; G3 has four response coordinates, only the first of which carries the skew signal.

What changes between two nearby histories? (schematic)

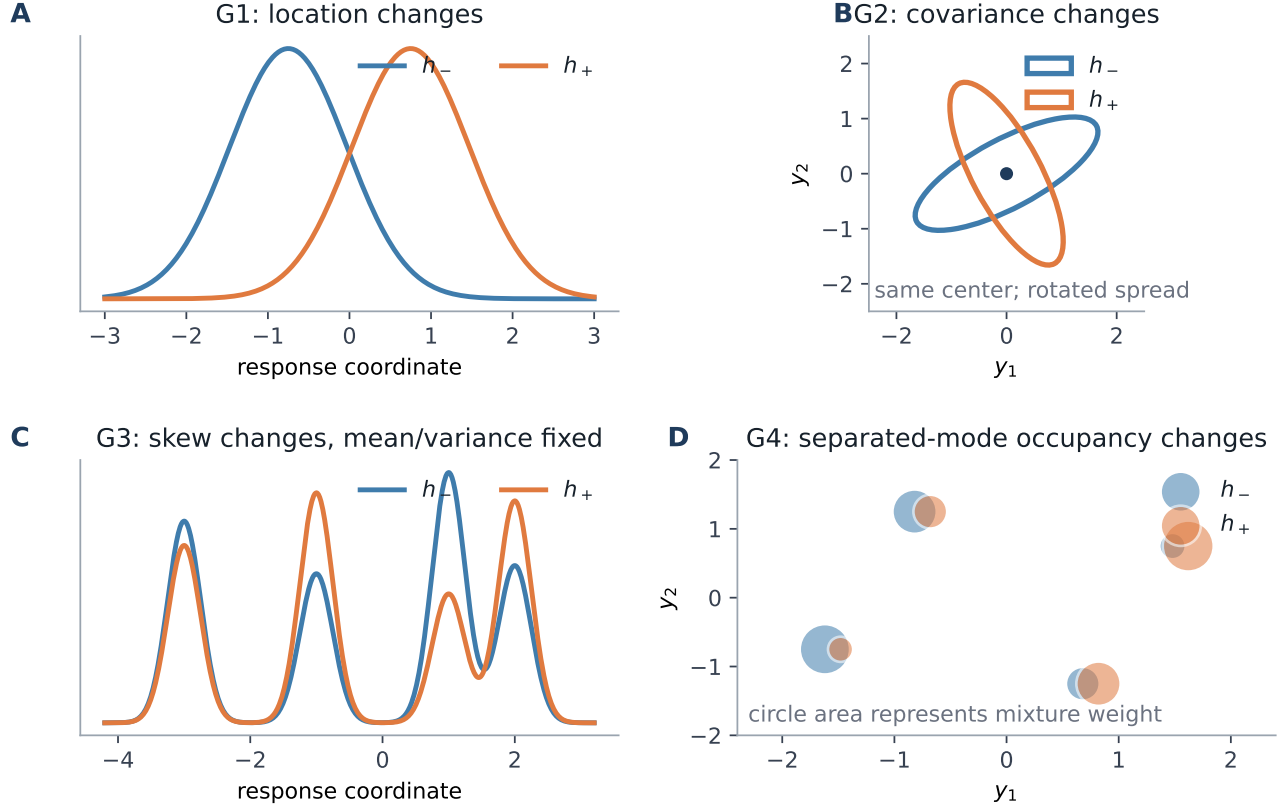


Figure 1: Schematic, not fitted data. Blue and orange show the laws at two nearby histories. The tests become progressively less visible to mean-only or Gaussian methods.

G1: nonlinear Gaussian location.

$$Y = m(h) + \varepsilon, \quad m(h) = W_2 \tanh(W_1 h), \quad \varepsilon \sim \mathcal{N}(0, \Sigma_0).$$

Only the mean changes. A mean network or Gaussian model is already correctly specified, so this is a sanity check rather than a case where flexible SID should win.

G2: history-dependent Gaussian covariance.

$$Y \mid h \sim \mathcal{N}\left(0.3m(h), R \operatorname{diag}\{\exp(b + 0.6 \tanh(Ah/\sqrt{8}))\} R^\top\right).$$

Spread and correlation change with history. A mean-only method is insufficient. The tested Gaussian baseline was diagonal after global standardization, so it was not the strongest possible full-covariance comparator.

G3: matched mean and variance, changing skew. The informative coordinate is a four-component Gaussian mixture with means $(-3, -1, 1, 2)$, component standard deviation 0.25, and weights

$$w(h) = \frac{1}{4} \mathbf{1} + 0.12 \tanh(a^\top h) \left(-0.2, \frac{2}{3}, -1, \frac{8}{15}\right).$$

The weight direction is constructed so the first two moments stay fixed. Only higher-order shape changes.

G4: separated mixture modes. Six isotropic Gaussian components have standard deviation 0.65 and separated centers. Their history-dependent weights are $w(h) = \text{softmax}(Bh)$ with $\text{rank}(B) = 2$. The signal is which mode becomes more likely, not merely a small mean shift.

4 Exactly how the frozen tests were run

4.1 Core predictive-law and clean-tangent panel

Item	Frozen value
Data split per case	8,000 training; 1,600 validation; 512 test examples
Independent axes	2 generator seeds $\times$ 2 data seeds $\times$ 2 model seeds = 8 cells per generator/backend
Optimization	CPU; AdamW; learning rate 10^{-3} ; weight decay 10^{-4} ; batch 256; at most 60 epochs; patience 10; gradient norm clipped at 1
Law evaluation	Fair multivariate energy score on 96 histories with 64 generated samples; exact NLL where the model was normalized
SID evaluation	64 tangent pairs; 12 evaluation histories; 48 oracle conditional draws per history; exact oracle density, score, tangent, and samples available
Core completion	784 of 784 model cases completed; no failed cases

4.2 Finite-contrast panel

The finite run reused the same fitted backends. It tested $\delta \in \{0.05, 0.12, 0.25\}$ under clean responses and standardized response noise $\sigma = 0.12$. The primary comparison in this guide is clean $\delta = 0.12$.

The balanced classifier used 768 training anchor histories and 384 independent calibration anchors, with one draw from each endpoint per anchor. Evaluation used 16 anchors and 64 draws per endpoint. Direct Riesz evaluation used 96 draws per side. The panel produced 212,832 valid metric rows and 4,704 diagnostic rows. There was one adapter seed, so the eight-cell variation shown in the figures comes from the frozen generator/data/model seeds, not repeated classifier initializations.

4.3 Metrics and baselines

NLL tests a normalized density. Lower is better. We report the paired median reduction relative to the heteroscedastic Gaussian.

Fair energy score tests generated samples without requiring a density. It compares a held-out response with an independent sample ensemble and corrects the finite-ensemble self-distance term. Lower is better.

NRMSE is root-mean-squared error normalized by the RMS size of the exact target:

$$\text{NRMSE}(\hat{f}, f) = \sqrt{\frac{\mathbb{E}(\hat{f} - f)^2}{\mathbb{E}f^2}}.$$

Thus 0 is perfect and 1 is exactly the error of predicting $\hat{f} = 0$.

Effect slope regresses the estimated typed effect on the exact effect through the origin. Slope 1 means the right magnitude; slope 0 means complete attenuation. **AUC** evaluates endpoint discrimination; 0.5 is chance.

Important limitation. The reported “best” route is the best median among methods already present in the frozen panel. This is descriptive model comparison, not a fresh confirmatory test of a single preregistered winner. The oracle and zero-effect baselines prevent us from calling an above-one NRMSE a success, but specialized external baselines still matter.

5 Results on the simple systems

5.1 Did the models learn the predictive law?

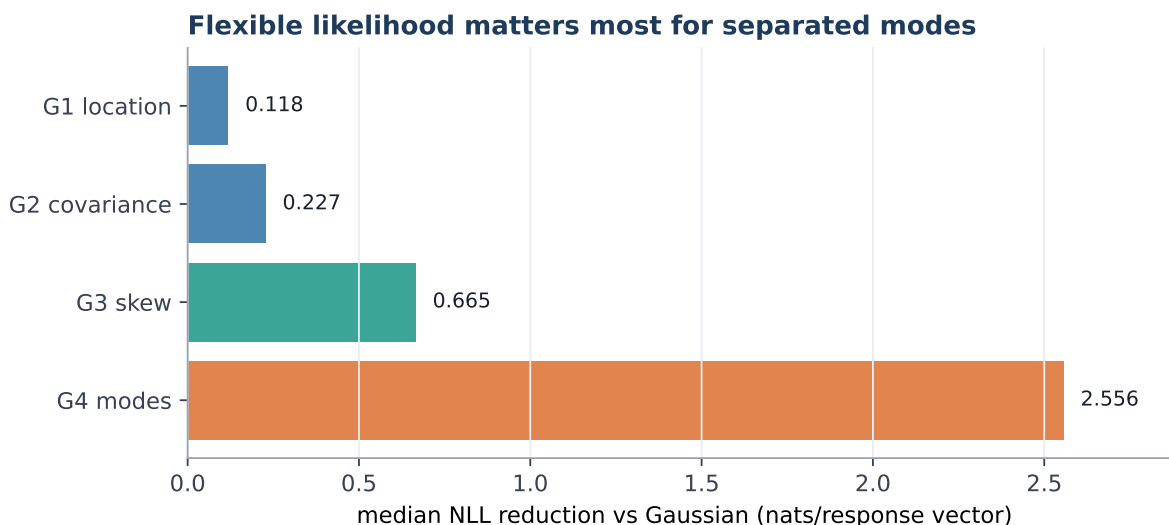


Figure 2: Median held-out NLL reduction relative to the tested heteroscedastic Gaussian. The winning normalized model was the eight-component autoregressive MDN in all four systems.

The law-fit result is modest for G1–G3 and much clearer for G4. The corresponding fair-energy improvements were only 0.41%, 0.66%, 0.44%, and 2.25% for G1–G4. This matters for interpretation:

- G1 is nearly Gaussian by construction. Extra flexibility buys essentially nothing practically.
- G2 is Gaussian but has structured covariance. The small gain partly reflects the weakness of a diagonal Gaussian baseline.
- G3’s MDN improves likelihood by 0.665 nats, so its non-Gaussian shape is learnable at the law level. That does *not* mean the history derivative of skew is learned.
- G4 is the first simple system where flexible law modeling is materially useful: separated modes give a 2.556-nat NLL gain.

5.2 Did the methods recover the history effect?

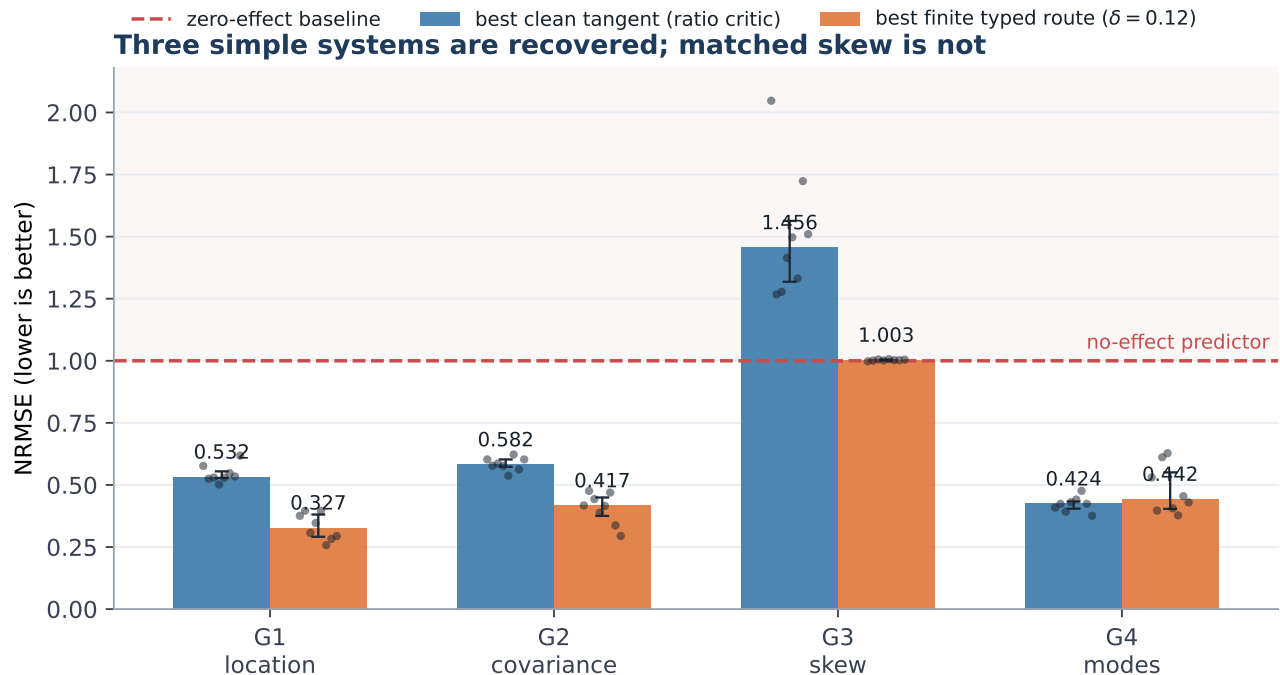


Figure 3: Bars are eight-cell medians; dots are the individual frozen cells and error bars are interquartile ranges. Blue uses the best clean infinitesimal route, which was the ratio critic in G1–G4. Orange uses the best clean finite typed route at $\delta = 0.12$. The routes differ because each column asks for the best estimator of its declared quantity.

G1, G2, and G4 clearly beat the zero-effect baseline. G3 does not. The precise finite winners were coupled anchored-EDM samples for G1, coupled flow-matching samples for G2, signed Riesz on legacy diffusion for G3, and the MDN central density ratio for G4. The G3 winner at 1.003 is a failure, not a tie worth celebrating.

5.3 Why diffusion looked better than it really was

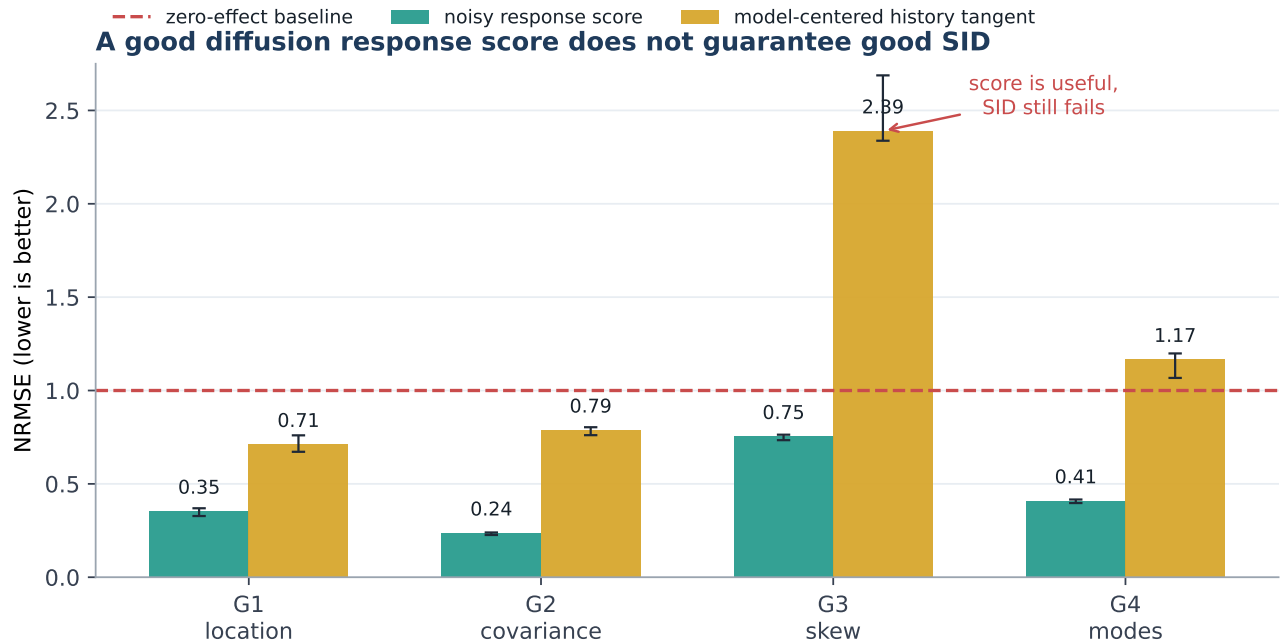


Figure 4: Legacy diffusion at standardized response noise $\sigma = 0.12$. A low response-score NRMSE evaluates the native denoising target. The model-centered history tangent is the implementable SID quantity. The two errors decouple sharply for skew and separated modes.

Legacy diffusion estimates the noisy response score reasonably well in all four systems. Yet its history tangent is worse than zero on G3 and G4. This is not mainly caused by the small response corruption: the exact clean-to-noisy tangent mismatch was only 0.050, 0.028, 0.094, and 0.002 NRMSE on G1–G4. The failure occurs when a good response score must be differentiated with respect to history and then centered using model samples.

Stabilized EDM, Gaussian anchoring, flow matching, and a fixed-anchor stochastic interpolant improve optimization or sampling in some cases, but they do not change the estimand. This is why better denoising loss or better sample energy cannot by itself establish SID validity.

5.4 System-by-system verdict

System	Verdict	What the experiments say	Best practical interpretation
G1	works	Tangent 0.532; finite typed 0.327. The effect is easy and stable.	Use a mean/Gaussian model unless a shared richer back-end is needed for other channels.
G2	partial	Tangent 0.582; finite covariance 0.417. History-dependent spread is recoverable.	Promising, but compare against full-covariance regression before claiming a new-method win.
G3	does not work	Tangent 1.456; finite skew 1.003 at $\delta = 0.12$. Better NLL did not translate to skew SID.	Current infinitesimal/dictionary approach has not solved matched skew.
G4	works	Tangent 0.424; finite typed 0.442; best finite witness 0.342; flexible likelihood gains 2.556 nats.	Strongest simple evidence for ratios and normalized finite SID on non-Gaussian mode occupancy.

At the larger G3 scale $\delta = 0.25$, direct EDM samples reached typed NRMSE 0.660 versus an oracle Monte Carlo floor of 0.430; at $\delta = 0.12$ the same route was about 1.04. This is a useful finite-signal hint, but it changes the perturbation scale and therefore the scientific estimand. It is not evidence that the clean infinitesimal skew problem has been solved.

6 The hard G8 check: a pure fourth-order path effect

G8 was designed to prevent easy methods from winning by construction. Its future path law changes the mixture weights of signed, rotated motif frames while keeping the conditional mean and covariance exactly fixed. The target readout is fourth order. The endpoint signal is real: the exact Bayes classifier has AUC 0.768.

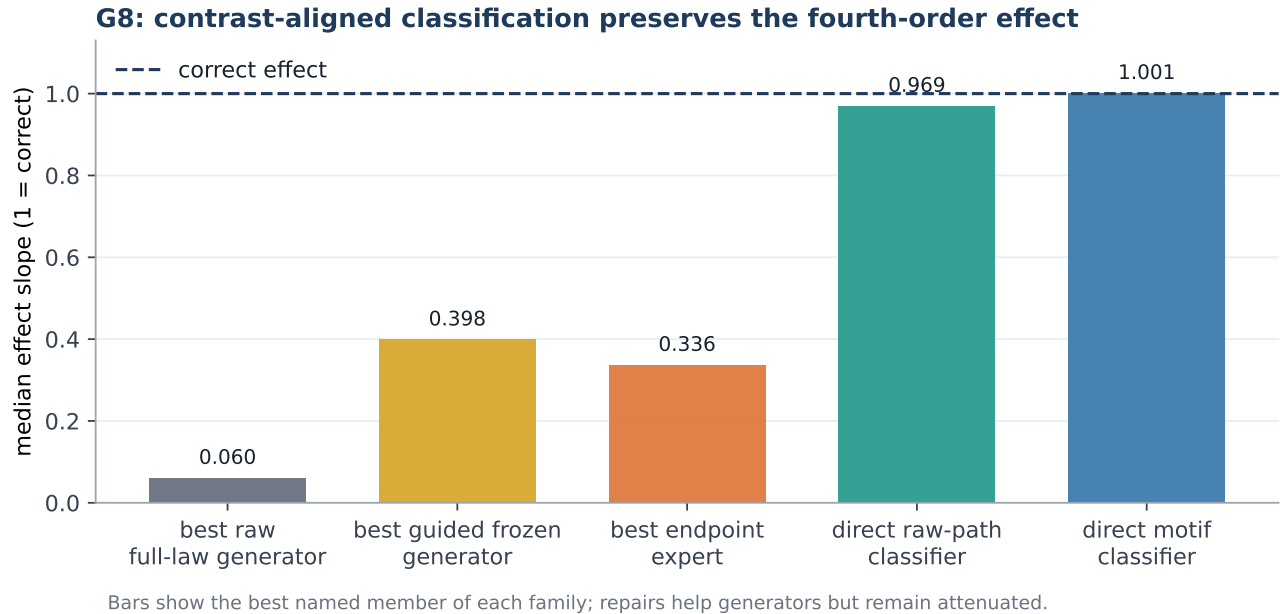


Figure 5: Median fraction of the true G8 typed effect recovered. The best raw full-law generator was the MDN (slope 0.060); the best guided frozen generator was EDM (0.398); the best endpoint expert was EDM (0.336). A direct classifier on the raw path recovers 0.969, and the motif-aligned classifier recovers 1.001.

The direct classifier result is not a consequence of leaking the answer through a low-order statistic. A linear motif classifier had AUC 0.499 and a quadratic motif classifier 0.508. Both recovered essentially none of the typed effect. In contrast:

Estimator	AUC	Witness NRMSE	Median typed-effect error
Motif MLP	0.768	0.229	1.13%
Raw path/history MLP	0.763	0.421	3.05%
Typed quartic logistic	0.768	0.372	1.16%
Quadratic motif	0.508	1.002	99.64%
Linear motif	0.499	1.000	100.00%

Classifier guidance proves that part of the missing effect is global mass allocation: it raises EDM from slope 0.010 to 0.398. Endpoint experts make the history split explicit but reach only 0.336. Exact rejection sampling repairs the bounded-energy score contract but leaves its G8 quartic NRMSE at 1.000. A fixed-anchor stochastic interpolant works on G1 yet has G8 slope near zero. Together these controls rule out the simple explanations “the signal is absent” and “the sampler is merely unstable.”

Bottom line. G8 is the strongest evidence for the core idea: a generic full-law objective may spend capacity on dominant within-mode geometry and ignore a weak but scientifically central history-controlled change in mode weights. Training directly on the declared contrast preserves that effect.

7 What works best, and when

Question	Preferred current method	Reason
Does the conditional mean change?	Mean MLP or Gaussian; optionally direct coupled samples	Correctly targeted, simple, and statistically efficient.
Does covariance change?	Direct covariance readout from coupled samples; ratio critic as a general screen	The typed moment is easier than a full witness. Add a full-covariance Gaussian baseline.
What is the clean local law tangent?	Ratio critic	Directly estimates the joint-versus-product log ratio whose history derivative equals the desired tangent.
What changes between two supported histories?	Balanced finite classifier or exact finite density ratio	Direct objective alignment and a bounded witness.
Which declared higher-order feature changes?	Direct typed difference or a contrast-aligned classifier	G8 shows this can succeed when full-law generators erase the feature.
Do we need realistic conditional samples?	MDN/Transformer, anchored EDM, or flow, chosen by held-out NLL/energy	These are predictive backends. Validate the relevant typed effect separately.
Do we need a changepoint score?	First obtain a validated finite typed effect, then place a calibrated sequential method around it	Changepoint inference is downstream of a trustworthy local contrast.

Most developed project idea. Finite, typed, support-aware SID is the most coherent method family. The finite endpoint classifier is its cleanest general estimator; direct typed sample contrasts are its efficient special case; normalized density ratios are valuable when one fitted law must answer many contrasts.

Best infinitesimal idea. The ratio critic is the best current clean-tangent route. It works on six of eight frozen laws overall, but its failures on G3 and support-concentrated G7 show that it is not universal.

Role of diffusion and flow. Keep them as backends when sampling, coupling, or complex within-mode geometry is valuable. Do not use native score error as proof of SID. Require a separate finite typed validation and agreement with a direct ratio/classifier baseline.

What has not worked. Generic score-to-history differentiation, the current signed Riesz dictionary, bounded-energy fitting, and fixed-anchor stochastic interpolation have not recovered the hard shape effects reliably. More training of the same objective is unlikely to fix an estimand mismatch.

Important limitation. The project has evidence of a useful estimator, not yet evidence of universal superiority. G1 has a simpler correct baseline; G2 still needs a strong full-covariance baseline; G3 is unsolved; G4 and G8 are the clearest places where the contrast-aligned approach adds something genuinely different.

One-sentence synthesis

Learn the predictive law when you need a predictive law, but train on the finite, typed history contrast when that contrast is the scientific target.

A Compact implementation details and provenance

A.1 Frozen core model settings

Model	Main frozen architecture
Mean / Gaussian / Gaussian DSM	Two hidden layers, width 64
Autoregressive MDN	Two hidden layers, width 56, eight Gaussian components per output
Autoregressive Transformer	$d_{\text{model}} = 48$, four heads, two Transformer layers, feedforward width 128, eight mixture components; only G5 and G8
Affine flow	Two hidden layers, width 64, six coupling layers, max log scale 1.5
Ratio critic	Two hidden layers, width 56, interaction rank 32
EDM / anchored EDM	Two hidden layers, width 64; noise 0.01–4.0; 18 sampling steps; $\rho = 7$; EDM $\sigma_{\text{data}} = 0.5$; anchored base-NLL weight 0.5
Flow matching / Gaussian-source flow	Two hidden layers, width 64; 18 Heun steps; Gaussian-source base-NLL weight 0.25
Bounded energy	Two hidden layers, width 64; tilt bound 1; Gaussian base-NLL weight 0.25

A.2 Audit boundaries

- Training, model selection, and evaluation were performed on exact synthetic systems. The oracle supplied exact densities, conditional samples, response scores, history tangents, and typed effects.
- The core seed product was frozen before inspection of the final results. Finite adapters used one adapter seed, so adapter-initialization uncertainty is not measured in the main finite table.
- The bounded-energy core fair-energy result was later invalidated because it used dependent finite-pool resampling. The exact rejection reevaluation fixed that sampling issue and still failed the G8 typed target. This guide does not use the invalid core energy number.
- “Best” always means best among tested methods on a common frozen estimand. It does not mean best among all possible statistical methods.

A.3 Frozen artifact locations

- Core configuration and results:
`history_tangent_benchmark/configs/stable_sid_core_20260713.yaml`
`history_tangent_benchmark/results/stable_sid_core_20260713/`
- Finite configuration and results:
`history_tangent_benchmark/configs/stable_sid_finite_20260713.yaml`
`history_tangent_benchmark/results/stable_sid_finite_20260713/`
- G8 direct classifier audit:
`analysis/g8_typed_classifier_benchmark_2026-07-13/`

- G8 classifier guidance and endpoint experts:
analysis/g8_classifier_guided_generator_2026-07-13/
analysis/g8_endpoint_expert_decomposition_2026-07-13/
- Figure-generation script for this guide:
reports/sid_simple_guide_2026-07-13/make_figures.py

A.4 How to read a result without being misled

Ask these questions in order:

1. What is the estimand: law fit, outcome score, clean tangent, finite witness, or typed effect?
2. What information did the method receive: a density, samples, endpoint labels, the perturbation direction, or a hand-chosen feature dictionary?
3. What is the relevant baseline: zero effect, oracle Monte Carlo, mean/Gaussian, a specialized moment model, or a two-sample classifier?
4. Is the number below NRMSE 1 in every seed cell, or only in the median?
5. Does the scientific claim match the perturbation scale δ and response noise level actually tested?